

Week 7 - Logistic Regression

- Classification Problems
- Odds, logits, Sigmoid function
- Confusion Matrix, Precision, Recall, Accuracy.

Classification Problem:

$Y \rightarrow$ No longer Continuous Variable
Now it is a categorical variable.

Regression $\rightarrow$ Continuous Variable
Classification $\rightarrow$ Categorical Variable.

It is a Supervised Learning Technique.

How Logistic Regression Works?

If we can get Probability somehow
then we can convert it into
a label using some predecided Threshold.

Since Prob. is number can we use $\ln$ (log)?

$\rightarrow$ Maybe Yes, but that is very restrictive
as the 'y' must always be 0 to 1.

Hence we need something which has range
 $-\infty$ to $+\infty$ & using that we get Prob.

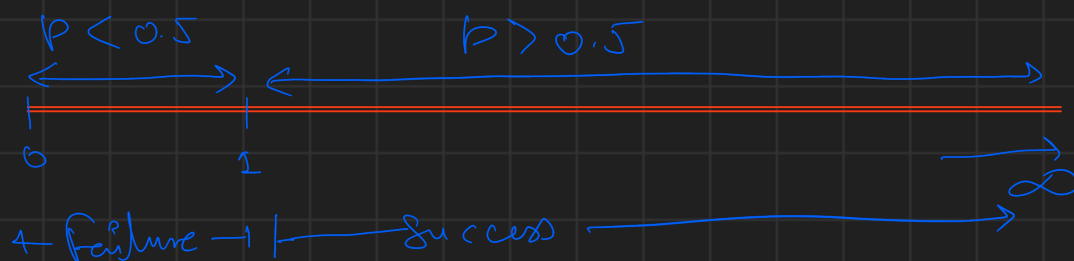
Odds:

$$\text{Odds} = \frac{\text{Prob of Success}}{\text{Prob of failure}}$$

if p is Prob of Success then

$$\text{Odds} = \frac{p}{1-p}$$

Notice Range of odds is 0 to ∞



Now if we take $\log$: $\text{Logit} = \log \text{ of odds}$

$\log(\text{odds}) \rightarrow$ Then Range become $-\infty$ to ∞

Therefore we use Linear Regression to model $\log(\text{odds})$

This is known as Logistic Regression

$\Rightarrow$ The objective here is to Maximise $\log$ Likelihood of odds.

Now,

$$Y = \log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

\$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

$$\Rightarrow \frac{p}{1-p} = \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)$$

$$\Rightarrow p = \frac{\exp(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}{1 + \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}$$

$$\Rightarrow p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

() probability of success.

Sigmoid or logistic function:

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

We use Sigmoid / logistic function to

convert the logit (log of odds)
to Probability.

Confusion Matrix:

		Predicted	
		+	-
Actual	+	2 True Positive	2 False Negative
	-	1 False Positive	1 True Negative
	+		
	-		
	+		
	-		

Accuracy:

$$\frac{TP + TN}{TP + FP + TN + FN} = \frac{\text{diagonal sum}}{\text{Total samples}}$$

Fraction (%) of correctly predicted samples out of total samples.

Precision

Fraction of correctly predicted samples out of the total samples that were predicted as that class

Recall

Fraction of correctly predicted samples out of the total samples that belonged to

$$\text{Precision}_{(+ve)} = \frac{TP}{TP + FP}$$

$$\text{Precision}_{(-ve)} = \frac{TN}{TN + FN}$$

$$\text{Recall}_{(+ve)} = \frac{TP}{TP + FN}$$

$$\text{Recall}_{(-ve)} = \frac{TN}{TN + FP}$$

		Predicted	
		+ve	-ve
Actual	+ve	True Positive	False Negative
	-ve	False Positive	True Negative